

CS60078: Complex Network Projects

Graph Based Methods for Predicting Drug-Target Binding Affinity

Midterm Report

Kovvada Dhyana Vardhan (21CS30030)

Tanishq Prasad (21CS30054)

Ayush Sunil Munot (21ME3FP55)

February 13, 2026

1 Problem Statement

A major challenge in computational drug discovery is predicting the binding capability of a drug with a specific protein, named as drug-target interaction (DTI) prediction. This domain has witnessed a growth in research interest, particularly for its application in drug repurposing or repositioning [20], especially after the world-wide SARS-CoV-2 outbreak. In the context of DTI [27], the primary objective is to quantify the binding affinity between FDA-approved drugs and novel target proteins. Machine learning methodologies are increasingly employed to predict these affinities. By systematically testing existing FDA-approved drugs for efficacy against new diseases, drug repurposing techniques can substantially shorten the traditional drug development timeline, while keeping the costs significantly lower.

2 Previous Solutions and Limitations

- **Traditional Methods:** A high throughput screening experiment examines bioactivities between drugs and proteins in this traditional approach, which makes it a costly and time-consuming process [3, 16]. This is an infeasible task as there are millions of drug-like compounds [6] and hundreds of potential targets.
- **Statistical and Machine Learning models:** Many statistical and machine learning models [9, 8, 4] were used to predict DTI (drug-target interaction) of FDA-approved drugs on new targets. The machine learning techniques, which have been previously used for DTI prediction, are affinity similarity (SimBoost) [7] and Kronecker regularized least squares based (KronRLS) [2, 1]. SimBoost [7] is a gradient boosting method that depends on the features extracted from the drugs, the targets, and the drug-target pairs. They use feature engineering to extract object-based and homogeneous as well as heterogeneous network-based features. High dependence on feature engineering can lead to missing out on information if those features are not well designed or if any patterns are missed. Later, it was seen that they were outperformed by the DL models.
- **Deep Learning Models:** Deep learning models predict better than the above-mentioned machine learning models.
 - **Earlier DL Models:** Some of the deep learning models proposed earlier for drug repurposing, are DeepDTA [27], WideDTA [26], MT-DTI [19], DeepCPI [21], GANs-DTA [24], Attention-DTA [25] and 1-D CNN [13]. In DeepDTA [27], drugs (provided in SMILES notation) and targets are represented as sequence of characters. In WideDTA [26], drugs and proteins are represented as words instead of characters as

in DeepDTA [27]. Similarly, other deep learning based models are DeepCPI [21], MT-DTI [19], Attention-DTA [25] and 1-D CNN [13], which are also used for drug-target interaction (DTI) prediction. Nevertheless, all of these methods convert drug compounds into corresponding string representations, which are not ideal to represent drug molecules. Using such strings may lead to exclusion of the topological information of the drug molecules. As a result, it decreases the performance of the above-discussed methods and their power of predicting drug-target affinity. However, Graph Neural Network (GNN) based works [12] [15] used topological information of drug molecules by exploiting the chemical structures of drugs as provided in SMILES notation. But, both [12] [15] used 1-D CNN to encode protein sequence information which are known to be inferior than the LSTM based encoders for protein sequencing.

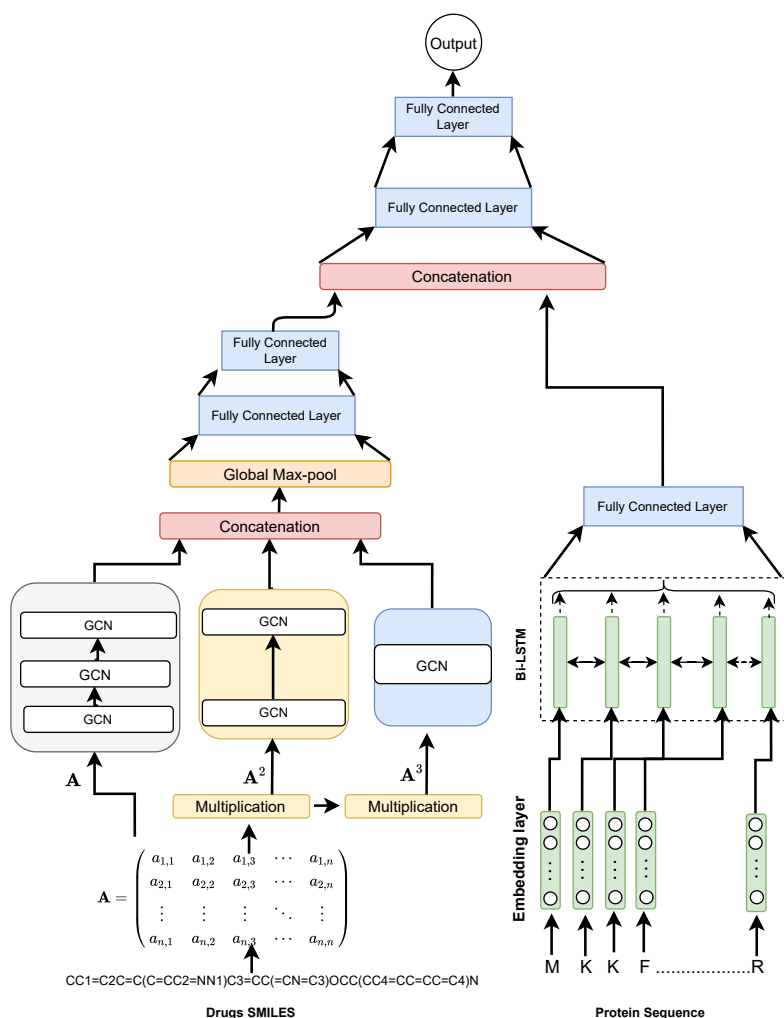


Figure 1: Model architecture of **DeepGLSTM**

3 Proposed Novel Solution (DeepGLSTM)

DeepGLSTM's architecture is shown in Fig. 1. **DeepGLSTM** has two functional modules. The first one helps to capture the topological information from the drug molecules. On the other hand, the second one captures the sequential information from the targeted protein structures.

- Graph Convolutional Network (GCN) Module:** They represent each drug compound by its corresponding SMILES (Simplified Molecular Input Line Entry System)[22] notation. This notation always considers the compounds as a graph of the interactions between atoms (i.e., the nodes). To describe a node feature, they use a set of atomic features adapted from DeepChem [18]. Here, each node is represented as a multi-dimensional binary feature vector. The feature vector of every node demonstrates five pieces of information (i.e., the atom symbol, the number of adjacent atoms, the number of adjacent hydrogens, the implicit valence of the atom and whether the atom is in an aromatic structure) [18]. Then they use the open-source chemical informatics software RDKit [11] to convert the SMILES data to its corresponding molecular graph representation (i.e., adjacency representation $A \in R^{N \times N}$), which helps them to extract the required atomic features. To represent the graph level features for every drug compound, incorporation of multi-hop connectedness relationship between the nodes present in a graph may turn out to be important. For this reason, they introduce the idea of using GCN in their architecture. They introduce three blocks consisting of GCNs. In the first block, they stack three GCN layers. In the second block, they stack two GCN layers and in the final block they use only one GCN layer. For every drug compound, they compute a simple propagation rule as mentioned in [10] by taking the adjacency representation ($A \in R^{N \times N}$) generated from the RDKit tool and a node feature matrix $X \in R^{N \times C}$ (where C is the number of features per node) as inputs in the first block. To overcome the degree normalization problem of the adjacency representation, they compute the normalized adjacency representation (A_{norm}) as in Eqn. 1, where $D \in R^{N \times N}$ is the degree matrix representation of A .

$$A_{norm} = D^{-1/2} A D^{-1/2} \quad (1)$$

The Eqn. 2 makes the first block workable, where W is the trainable weight, $H_e^0 = X$ is the i^{th} layer output representation and σ is a non-linear activation function. Thus, i^{th} layer of the GCN module produces a global representation ($H_e^i \in R^{N \times M}$) for A .

$$H_e^i = \sigma(A_{norm} H_e^{(i-1)} W^{(i-1)}) \quad (2)$$

To capture local reachability and structural information beyond immediate neighbors, the subsequent blocks utilize the concept of power graphs. Specifically, the second GCN block incorporates the squared adjacency matrix (A^2), which effectively connects every node to its neighbors’ neighbors (i.e., nodes at a shortest-path distance of 2). Following the same methodology established in the first block, they apply degree normalization to this squared representation. This allows the model to generate a global representation that encapsulates 2-hop dependencies.

Similarly, the third GCN block extends this scope by utilizing the cubic adjacency matrix (A^3). This block applies the identical degree normalization and propagation mechanisms to capture 3-hop connectivity information. Finally, to integrate these output representations from all three blocks, corresponding to 1-hop, 2-hop, and 3-hop neighborhoods, are concatenated. This unified feature set is subsequently passed through a global max-pooling layer and fully connected layers to arrive at a graphical representation of the molecule.

- Bi-LSTM Module:** Similarly, for protein sequences, one could also represent them as graphs. But, doing so is more difficult because the tertiary structure of a protein is not always available in a reliable form. So, for protein sequences they use the popular one-hot encoding representation [15].

The protein sequence is a string of ASCII characters, which represents amino acids. They initialize the protein sequence embedding layer by mapping every amino acid to an English letter and subsequently, by marking the amino acid with the sequential index of that letter

in the English alphabet. They pass the embedding representation ($c \in \mathbb{R}^{d_p}$, where d_p is the dimension of the protein sequence embeddings) to a Bi-LSTM layer, which captures the dependencies between the characters in a sequence ($p = [c_1, c_2 \dots c_n]$) of length n . They get the output representation $h_t \in \mathbb{R}^{2d_l}$, where d_l represents the number of output units used in each LSTM cell. They compute Eqn. 3 for execution of the LSTM.

$$\begin{aligned}\vec{h}_t &= \overrightarrow{LSTM}(c_t, h_{t-1}) \\ \overleftarrow{h}_t &= \overleftarrow{LSTM}(c_t, h_{t+1}) \\ h_t &= \vec{h}_t || \overleftarrow{h}_t\end{aligned}\tag{3}$$

Then they concatenate the output representations of the Bi-LSTM and the fully connected layers. Finally, they feed the concatenated representation into some fully connected layers followed by the output layer to predict the affinity score.

4 Datasets

Dataset	# of drugs (compounds)	# of targets proteins	Total number of drug-target pairs used
Davis(k_d)	68	442	30056
KIBA	2111	229	118254
DTC(pk_i)	5983	118	67894
Metz(pk_i)	1471	170	35307
ToxCast(AC_{50})	7657	328	342869
STITCH	724471	15258	1244420

Table 1: Dataset statistics

In the field of drug-target interaction (DTI) prediction, several benchmark datasets are utilized to validate computational models. These datasets primarily consist of drug compounds (represented as SMILES strings), target proteins (represented as amino acid sequences), and their corresponding binding affinity scores. The following section details four prominent datasets used in this domain: Davis, KIBA, DTC, and Metz.

4.1 Dataset Details

- **Davis Dataset:** This dataset focuses on the selectivity of kinase inhibitors. It contains interaction data for **68** drugs and **442** target proteins, creating a dense matrix of roughly 30,056 interaction pairs. The binding affinity is measured using the dissociation constant (K_d), typically reported in nanomolars (nM). A lower K_d value indicates higher binding affinity.
- **KIBA Dataset:** The Kinase Inhibitor BioActivity (KIBA) dataset addresses the challenge of heterogeneous data sources by integrating various bioactivity types, such as K_i , K_d , and IC_{50} , into a single, unified scoring metric known as the KIBA score. It comprises **2,111** drugs and **229** targets, totaling 118,254 interactions. Unlike standard inhibition constants, a higher KIBA score typically indicates a higher binding affinity in processed datasets.
- **DTC (Drug Target Commons):** DTC is a community-driven open-data platform designed to crowd-source and standardize drug-target bioactivity data. The subset often used for benchmarking includes **5,983** drugs and **118** targets with approximately 67,894 interaction pairs. The primary affinity metric for this subset is the negative logarithm of the inhibition constant (pK_i).

- **Metz Dataset:** The Metz dataset is another kinase-focused collection, specifically targeting tyrosine kinase inhibitors. It includes **1,471** drugs and **170** targets, providing 35,307 interaction data points. Similar to DTC, the binding affinity is often standardized to pK_i values.

4.2 Data Structure and Examples

To illustrate the format of these datasets, Table 2 provides representative examples of the input features (Drug SMILES and Protein Sequences) and their corresponding affinity labels.

Table 2: Representative Examples from DTI Datasets

Dataset	Drug (SMILES)	Target (Sequence Snippet)	Affinity
Davis	CC1=C...N=C1	MKK...LSR	K_d : 0.1 nM
KIBA	O=C(C)Nc1...	MST...KIG	Score: 12.1
DTC	CN(C)...C1	MEL...GDF	pK_i : 7.5
Metz	C0c1...cc2	MPR...AAV	pK_i : 6.2

Note: Sequences and SMILES are truncated for brevity. K_d represents the dissociation constant, while pK_i represents the negative log of the inhibition constant.

5 Results and Conclusion

Experimental evaluation confirms that DeepGLSTM outperforms existing state-of-the-art methods (including GraphDTA and DeepDTA) across multiple datasets (Davis, KIBA, DTC, Metz, and ToxCast), consistently achieving lower Mean Squared Error (MSE) and higher Concordance Index (CI) scores [14]. The ablation studies further highlight the critical contribution of multi-hop neighbor information via power graphs and the superiority of Bi-LSTM over 1-D CNNs for protein sequence encoding. In a practical application for SARS-CoV-2, the model screened 2,304 FDA-approved drugs against five viral proteins using a Combined Score (C_b), successfully identifying 18 potential repurposing candidates (such as Isoproterenol and Lactulose) and demonstrating the framework’s efficacy for rapid drug discovery [14].

6 Experimental Results and Analysis

6.1 Github Repository Setup

The code is publicly available and includes modifications from the original version to enable functionality and support ablation studies. Further details about these changes are provided in Task T3.

- **Link to Fork:** <https://github.com/ayush1298/DeepGLSTM>
- **Original Repository:** <https://github.com/shrimonmuke0202/DeepGLSTM>

6.2 Reproduction of Results

We aimed to reproduce the results and run the experiments described in the paper. To achieve this, we first implemented several key modifications to the original codebase to ensure all experiments and ablation studies could be successfully executed.

Key Codebase Modifications

We introduced three major categories of improvements to modernize the codebase, enable scientific reproduction, and ensure stability across environments.

1. Core Infrastructure & Modernization

PyTorch 2.x Upgrade and Dependency Standardization: We migrated the project from PyTorch 1.9 to PyTorch $\geq 2.0.0$ and PyTorch Geometric $\geq 2.3.0$. We replaced deprecated packages (e.g., `rdkit-pypi`) with official distributions and upgraded requirements to ensure compatibility with current hardware and Kaggle environments.

2. Model Architecture Flexibility

Dynamic GCN Blocks: We modified the `GCNNet` model architecture to support flexible layer configurations. The model now accepts a `layer_configs` argument, enabling dynamic activation or deactivation of specific Graph Convolutional blocks (Input A , A^2 , or A^3). This was necessary to isolate the contribution of each power graph component for ablation studies.

3. Reproducibility & Usability

Automated Ablation Studies: We created a specialized script to replicate ablation studies from the original paper (Table 5 and Table 6). This automatically trains multiple model variants (e.g., “Only 1st Block”, “DeepGLSTM Full”) and tabulates the Mean Squared Error (MSE) results for direct comparison.

Enhanced Codebase: We implemented a subset debugging mode for efficient testing on large datasets and improved data loading logic to support running with a data subset. Additionally, we added specialized logging for better organization of plots and results generation.

Implementation Overview

We implemented the novel solution proposed in the paper. This involved training the model from scratch and subsequently performing inference. We attempted to reproduce the results for four out of the six datasets mentioned in the paper, using a modified configuration tailored to our available computational resources. The specific details of this setup are provided in next section.

6.3 Quantitative Analysis

For the reproducibility experiments, we utilized a subset of the dataset (details of which are provided below). We also reduced the number of training epochs to 400 from the 1000 specified in the original paper, as running the full 1000 epochs was computationally prohibitive. The standard configuration from the paper was otherwise maintained, including a learning rate of 0.0005 and a batch size of 128.

Datasets Used for Reproduction

We elected to use a subset of approximately 30% of the original samples for evaluation. Due to the substantial size of datasets like DTC (pK_i), ToxCast (AC_{50}), and STITCH, we omitted them, as loading such large datasets was not feasible.

Table 3: Dataset Sampling for Reproducibility

Dataset	Total Samples (Original)	Samples Used (Reproduction)
Davis (K_d)	30,056	10,000
KIBA	118,254	10,000
DTC (pK_i)	67,894	10,000
Metz (pK_i)	35,307	12,000
ToxCast (AC_{50})	342,869	Not Used
STITCH	1,244,420	Not Used

Performance Metrics Comparison

The following table shows the experimental results reported in the original paper compared to the results obtained during our reproducibility study. We compare the key performance metrics: Mean Squared Error (MSE), Concordance Index (CI), and r_m^2 .

Table 4: Performance Metrics Comparison (Original vs. Reproduced)

Dataset	MSE		CI		r_m^2	
	Paper	Repro.	Paper	Repro.	Paper	Repro.
Davis	0.232	0.387	0.895	0.832	0.680	0.479
KIBA	0.133	0.382	0.897	0.768	0.792	0.412
DTC	0.149	0.477	0.895	0.757	0.841	0.533
Metz	0.294	0.410	0.810	0.767	0.640	0.516

Analysis of Results:

- **Performance Gap:** As expected, the reproduced MSE values are higher (worse) and CI values are lower than those in the original paper. For instance, on the Davis dataset, the MSE increased from 0.232 to 0.387.
- **Impact of Data Volume:** The increase in error is primarily attributed to the reduction in training data (only 30% used) and training duration. Deep learning models like LSTMs and GCNs are highly data-hungry; reducing the dataset size significantly limits the model’s ability to generalize, leading to lower Concordance Indices (CI).
- **Consistency:** Despite the performance drop, the model successfully converged and learned predictive patterns, achieving a CI of 0.832 on the Davis dataset. This indicates a strong ability to rank drug-target pairs correctly even with limited data.

Ablation Studies

1. Effectiveness of different components of DeepGLSTM This experiment demonstrates the effectiveness and contribution of different components of the DeepGLSTM model using the Davis Dataset (pK_d). We compared the full DeepGLSTM model against variants with fewer GCN blocks.

Table 5: Ablation Study: Model Components (Davis Dataset)

Exp.	Model Variant	MSE (Paper)	MSE (Repro.)
R1	1st block + 2nd block (2 layers)	0.237	0.438
R2	1st block + 2nd block $\rightarrow$ DeepGLSTM	0.232	0.429
R3	DeepGLSTM + Extra Layers	0.238	0.445

The results indicate that the third block of GCN with Bi-LSTM contributes most significantly to the model’s performance. Furthermore, the findings suggest that increasing the exponent of the power graph input beyond a certain value (e.g., a distance of 3 for shortest paths) may not significantly increase the model’s effectiveness, as it may not substantially change node reachability in drug graph structures.

- **Result:** The full DeepGLSTM model achieved the lowest MSE (0.429) compared to the smaller “1st + 2nd block” variant (0.438) and the deeper/larger variant (0.445).
- **Conclusion:** This confirms the paper’s finding that the specific 3-block architecture is optimal. Adding too many layers (R3) caused performance degradation, likely due to overfitting on the smaller subset of data. Our reproducibility results follow the same pattern of MSE as those in the original paper ($R2 < R1 < R3$), although the Mean Squared Error (MSE) values are higher due to running the models for only 100 epochs.

2. Effectiveness of using the Power Graph The effectiveness of the power graph representation was evaluated through three ablation experiments using DeepGLSTM on the Davis Dataset (pK_d). Each experiment isolated one of the three GCN blocks:

- **1st block:** Used only the adjacency matrix (A) as input.
- **2nd block:** Used only the squared adjacency matrix (A^2) as input.
- **3rd block:** Used only the cubic adjacency matrix (A^3) as input.

Table 6: Ablation Study: Power Graph Representations

Model	Input to GCN Block	MSE (Paper)	MSE (Repro.)
1st block GCN	A	0.239	0.442
2nd block GCN	A^2	0.245	0.443
3rd block GCN	A^3	0.244	0.467
DeepGLSTM	A, A^2, A^3	0.232	0.459

The Mean Squared Error (MSE) from these models indicates that the blocks utilizing A^2 and A^3 contribute to capturing complex topological features. Note that the reproduced MSE values are higher than the paper’s due to the reduced training to 100 epochs.

- **Result:** The 1st Block (using A) achieved an MSE of 0.442, while the 3rd Block (using A^3) had a higher MSE of 0.467.

- **Deviation:** Unlike the original paper, where combining all power graphs yielded significant improvements, our reproduction showed that the combined DeepGLSTM model (0.459) did not significantly outperform the single-block baselines in this specific 100-epoch low-data regime. This suggests that the complex multi-hop correlations captured by A^2 and A^3 require more training data to be learned effectively.

6.4 Qualitative Analysis

We visually compared the “Predicted vs. Measured Binding Affinity” scatter plots from the original paper against our reproduced results for the Davis, KIBA, and Metz datasets. We show these in Table 7.

General Trend Across all three datasets, our reproduced models successfully captured the positive correlation between predicted and actual binding affinity. The data points cluster along the diagonal ($y = x$) line, confirming that the model has learned the fundamental relationship between drug-target features and binding scores.

- **Davis Dataset:**

- *Paper:* Shows a tight, linear clustering with minimal variance.
- *Reproduction:* While the linear trend is visible, there is a noticeably wider “spread” or variance in the predictions, particularly for lower affinity values (5.0–6.0). This visual noise correlates directly with the higher MSE (0.387) we observed, likely due to the reduced training set size (30%).

- **KIBA Dataset:**

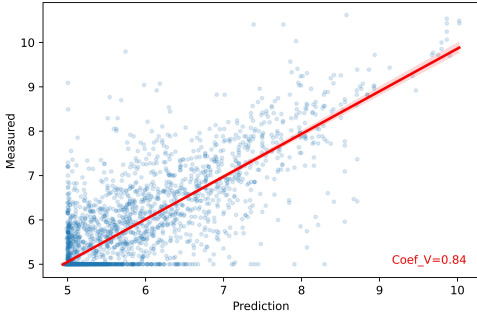
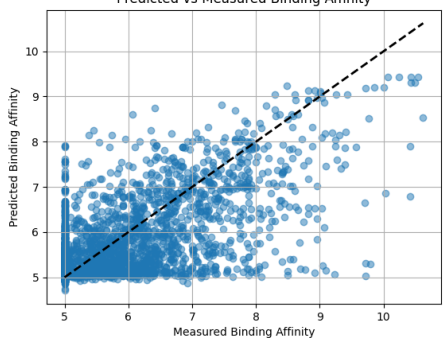
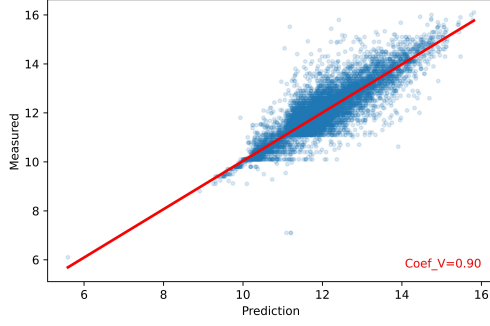
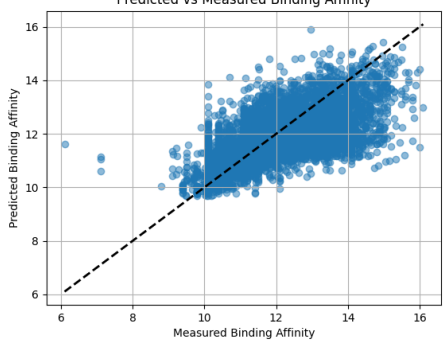
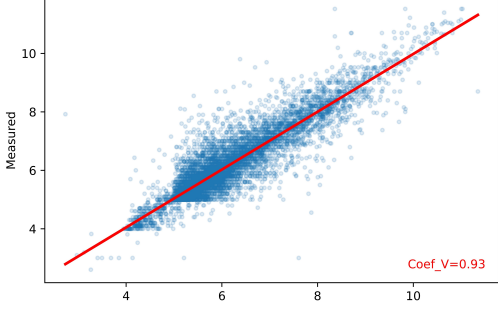
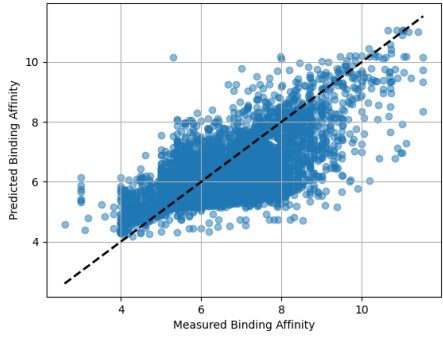
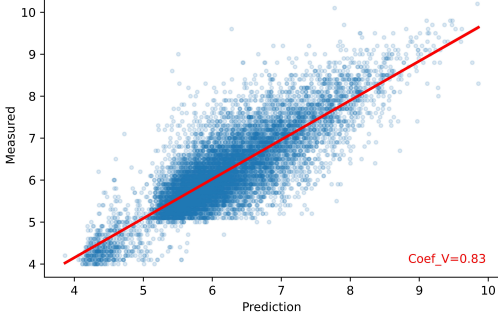
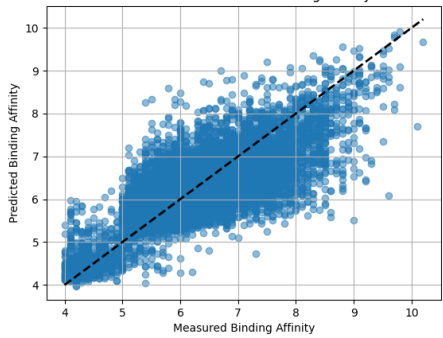
- *Paper:* Exhibits a very sharp, dense diagonal alignment.
- *Reproduction:* The reproduction is surprisingly robust, showing a dense concentration of points along the diagonal similar to the original. However, we observe some “saturation” where the model struggles to predict the most extreme high-affinity values (points falling slightly below the diagonal at the top right), which reflects the limit of training on fewer epochs (400 vs 1000).

- **Metz Dataset:**

- *Paper:* Shows a uniform distribution with a clear slope.
- *Reproduction:* The reproduced plot captures the slope but exhibits a “floor effect” or clustering at the lower end of the affinity scale. The cloud of points is broader than the original, indicating less precision in distinguishing between subtle differences in binding affinity for this specific dataset.

Outliers There are noticeable outliers in the reproduced plots where the model under-predicts high-affinity values (points below the diagonal). This “saturation” effect suggests the model, limited by 400 epochs, has not yet fully optimized its weights to capture the most extreme binding interactions.

Table 7: Visual Comparison: Predicted vs. Measured Binding Affinity

Dataset	Plot from Paper	Plot from Reproduction
Davis	 Scatter plot for Davis dataset showing Measured vs Prediction binding affinity. A red regression line is shown. Coef_V=0.84.	 Scatter plot for Davis dataset showing Predicted vs Measured Binding Affinity. A dashed diagonal line is shown.
KIBA	 Scatter plot for KIBA dataset showing Measured vs Prediction binding affinity. A red regression line is shown. Coef_V=0.90.	 Scatter plot for KIBA dataset showing Predicted vs Measured Binding Affinity. A dashed diagonal line is shown.
DTC	 Scatter plot for DTC dataset showing Measured vs Prediction binding affinity. A red regression line is shown. Coef_V=0.93.	 Scatter plot for DTC dataset showing Predicted vs Measured Binding Affinity. A dashed diagonal line is shown.
Metz	 Scatter plot for Metz dataset showing Measured vs Prediction binding affinity. A red regression line is shown. Coef_V=0.83.	 Scatter plot for Metz dataset showing Predicted vs Measured Binding Affinity. A dashed diagonal line is shown.

7 Improvements and Future Work

While DeepGLSTM demonstrates strong predictive capabilities, several avenues exist to enhance its architectural design and generalization power. Based on recent advancements in the field, we propose the following improvements:

- **Advanced Protein Representation (3D Structures & Pre-training):** The current model relies on 1D sequence data (one-hot encoding), which often fails to capture crucial tertiary structural information required for binding. With the advent of AlphaFold, we can generate reliable 3D structures from sequences and utilize Graph Neural Networks (GNNs) on these protein graphs. Furthermore, replacing the Bi-LSTM with pre-trained encoder-decoder models (e.g., ESM) can significantly improve the capture of long-range dependencies in protein sequences, as demonstrated in the LEP-AD framework [5].
- **Cross-Attention Mechanisms for Interpretability:** Currently, drug and protein embeddings are simply concatenated, which treats features independently before the fully connected layers. Introducing **Cross-Attention** mechanisms would allow the model to dynamically weight the interaction between specific drug atoms and proteins. This would provide interpretability regarding which substructures drive the binding affinity.
- **Enhanced Graph Learning (GAT/GIN):** The standard GCN layers treat neighbors relatively equally (modulo degree normalization). Incorporating Graph Attention Networks (GAT) or Graph Isomorphism Networks (GIN) would enable the model to learn the disproportionality in the drug structure, as some atoms influence binding significantly more than others [23].
- **Subgraph Extraction and Structural Invariance:** To improve generalization and convergence speed, the model could be augmented to extract subgraphs based on their importance to binding affinity. Learning these structural patterns and invariances allows the model to focus on the active pharmacophores rather than the entire molecule. This approach has been successfully implemented in SubMDTA [17].
- **Multi-scale Modeling:** Combining k -mer embeddings with parallel CNN and Bi-LSTM architectures can help capture both local motifs (via CNN) and global dependencies (via LSTM). This multi-scale approach ensures that neither short-term structural features nor long-term sequence contexts are lost [17].
- **Integration of 3D Docking Information:** Traditional auto-docking software often projects 3D interactions into 2D space (e.g., using spherical coordinates), leading to significant information loss. Future iterations could integrate raw 3D coordinate data directly using E(n)-equivariant GNNs to preserve geometric integrity without projecting to lower dimensions.

Contributions

- **Kovvada Dhyana Vardhan (21CS30030)**
 - Research for future work
 - Making Presentation
- **Tanishq Prasad (21CS30054)**
 - Writing the Report
 - Executing results for KIBA dataset

- Working on Improvements
- Refactoring Presentation
- **Ayush Sunil Munot (21ME3FP55)**
 - Codebase Refactoring
 - Reproduction of Results
 - Conducting Ablation Studies
 - Refactoring Presentation

References

- [1] A. Cichonska, T. Pahikkala, S. Szedmak, H. Julkunen, A. Airola, M. Heinonen, T. Aittokallio, and J. Rousu. Learning with multiple pairwise kernels for drug bioactivity prediction. *Bioinformatics*, 34:i509–i518, 2018.
- [2] A. Cichonska, B. Ravikumar, E. Parri, S. Timonen, T. Pahikkala, A. Airola, K. Wennerberg, J. Rousu, and T. Aittokallio. Computational-experimental approach to drug-target interaction mapping: a case study on kinase inhibitors. *PLoS Computational Biology*, 13:e1005678, 2017.
- [3] P. Cohen. Protein kinases—the major drug targets of the twenty-first century? *Nature Reviews Drug Discovery*, 1:309–315, 2002.
- [4] S. Corsello, J. Bittker, Z. Liu, J. Gould, P. McCarren, J. Hirschman, S. Johnston, A. Vrcic, B. Wong, M. Khan, et al. The drug repurposing hub: a next-generation drug library and information resource. *Nature Medicine*, 23:405–408, 2017.
- [5] Anuj Daga, Nawaz Khan, et al. Lep-ad: Language embedding of proteins and to drugs. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) Workshops*, pages 1–10, 2023.
- [6] M. Deshpande, M. Kuramochi, N. Wale, and G. Karypis. Frequent substructure-based approaches for classifying chemical compounds. *IEEE Transactions On Knowledge And Data Engineering*, 17:1036–1050, 2005.
- [7] T. He, M. Heidemeyer, F. Ban, A. Cherkasov, and M. Ester. Simboost: a read-across approach for predicting drug–target binding affinities using gradient boosting machines. *Journal Of Cheminformatics*, 9:1–14, 2017.
- [8] M. Iskar, G. Zeller, X. Zhao, V. Noort, and P. Bork. Drug discovery in the age of systems biology: the rise of computational approaches for data integration. *Current Opinion In Biotechnology*, 23:609–616, 2012.
- [9] S. Kinnings, N. Liu, P. Tonge, R. Jackson, L. Xie, and P. Bourne. A machine learning-based method to improve docking scoring functions and its application to drug repurposing. *Journal Of Chemical Information And Modeling*, 51:408–419, 2011.
- [10] T. Kipf and M. Welling. Semi-supervised classification with graph convolutional networks. In *5th International Conference On Learning Representations, ICLR 2017, Toulon, France, April 24-26, 2017, Conference Track Proceedings*, 2017.
- [11] G. Landrum. Rdkit documentation. *Release*, 1:4, 2013.

- [12] X. Lin, K. Zhao, T. Xiao, Z. Quan, Z. Wang, and P. Yu. Deepgs: Deep representation learning of graphs and sequences for drug-target binding affinity prediction. *ArXiv Preprint ArXiv:2003.13902*, 2020.
- [13] S. Majumdar, S. Nandi, S. Ghosal, B. Ghosh, W. Mallik, N. Roy, A. Biswas, S. Mukherjee, S. Pal, and N. Bhattacharyya. Deep learning-based potential ligand prediction framework for covid-19 with drug-target interaction model. *Cognitive Computation*, pages 1–13, 2021.
- [14] Shrimon Mukherjee, Madhusudan Ghosh, and Partha Basuchowdhuri. Deepglstm: Deep graph convolutional network and lstm based approach for predicting drug-target binding affinity. *arXiv preprint arXiv:2201.06872*, 2022.
- [15] T. Nguyen, H. Le, T. Quinn, T. Nguyen, T. Le, and S. Venkatesh. Graphdta: Predicting drug-target binding affinity with graph neural networks. *Bioinformatics*, 37:1140–1147, 2021.
- [16] M. Noble, J. Endicott, and L. Johnson. Protein kinase inhibitors: insights into drug design from structure. *Science*, 303:1800–1805, 2004.
- [17] Shourun Pan and Fei Xia. Submdta: drug-target affinity prediction based on substructure extraction and multi-scale protein representation. *BMC Bioinformatics*, 24(1):1–18, 2023.
- [18] B. Ramsundar, P. Eastman, P. Walters, and V. Pande. *Deep learning for the life sciences: applying deep learning to genomics, microscopy, drug discovery, and more*. O’Reilly Media, Inc., 2019.
- [19] B. Shin, S. Park, K. Kang, and J. Ho. Self-attention based molecule representation for predicting drug-target interaction. In *Machine Learning For Healthcare Conference*, pages 230–248, 2019.
- [20] S. Strittmatter. Overcoming drug development bottlenecks with repurposing: old drugs learn new tricks. *Nature Medicine*, 20:590–591, 2014.
- [21] M. Tsubaki, K. Tomii, and J. Sese. Compound-protein interaction prediction with end-to-end learning of neural networks for graphs and sequences. *Bioinformatics*, 35:309–318, 2019.
- [22] D. Weininger. Smiles, a chemical language and information system. 1. introduction to methodology and encoding rules. *Journal Of Chemical Information And Computer Sciences*, 28:31–36, 1988.
- [23] Fuyuan Xu and Gaoxi Xiao. Drug-target binding affinity prediction by graph neural network with attention mechanism. *Biomolecules*, 14(3):315, 2024.
- [24] L. Zhao, J. Wang, L. Pang, Y. Liu, and J. Zhang. Gansdta: predicting drug-target binding affinity using gans. *Frontiers In Genetics*, 10:1243, 2020.
- [25] Q. Zhao, F. Xiao, M. Yang, Y. Li, and J. Wang. Attentiondta: prediction of drug-target binding affinity using attention model. In *2019 IEEE International Conference On Bioinformatics And Biomedicine (BIBM)*, pages 64–69, 2019.
- [26] H. Öztürk, E. Ozkirimli, and A. Özgür. Widedta: prediction of drug-target binding affinity. *ArXiv Preprint ArXiv:1902.04166*, 2019.
- [27] H. Öztürk, A. Özgür, and E. Ozkirimli. Deepdta: deep drug-target binding affinity prediction. *Bioinformatics*, 34:i821–i829, 2018.